Probabilistic Methods

Lecture Notes

A clear route from counting and
events to random variables and data.

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Contents

Chapter 1

Introduction

These notes introduce probabilistic methods as a way of reasoning about uncertainty. The course begins with sample spaces and events, then develops counting techniques, probability distributions, standard distribution functions, conditional probability, and finally descriptive statistics based on generated data.

The central aim is to connect formal probability with interpretation. A probability calculation should not be treated as a mechanical substitution into a formula. It should begin with a clear description of the random experiment, the elementary outcomes, the events or random variables of interest, and the assumptions that make a chosen model appropriate.

1.1 Structure of the Course

These lecture notes are organized as a gradual construction of probabilistic thinking. The aim is not to begin with formulas and then search for examples, but to develop the mathematical language needed to describe uncertainty in a precise way. We start from the problem of counting and representing possible outcomes, then move toward events, probability, random variables, probability laws, empirical data, sampling, limit theorems, and statistical inference.

The course begins with combinatorics because many probabilistic models depend on the ability to count possible outcomes correctly. At this stage probability itself is not yet needed. What is needed is the more elementary skill of recognizing what kind of object an outcome is: a sequence, a selection, an arrangement, a choice with or without repetition, or an object whose elements may or may not be distinguishable. This first step builds the counting language that later allows us to describe sample spaces and favorable outcomes with precision.

The second stage has a special conceptual role. It is devoted to discovering the formal language of probability theory. The purpose is to show how a mathematical theory can be built from careful thinking, analysis, and abstraction. We examine how statements about a random situation select sets of outcomes, how these sets become events, how logical operations correspond to set operations, and why a numerical theory of probability should be placed on such a structure. This part is not primarily computational; it is meant to teach how probabilistic language is formed.

After this conceptual preparation, the notes turn to sample spaces and elementary probability. Several canonical examples are developed from the beginning: one first understands the experiment, decides what should count as an elementary outcome, constructs the sample space, defines events, and only then assigns probabilities. Different representations of the same structure, such as lists, tables, trees, and symbolic notation, are used to show that a sample space is an abstract mathematical model rather than a collection of observed data.

The next step is conditional probability. Here the central idea is that information changes the space in which we reason. Conditional probability is introduced by asking what it means to restrict attention to the cases in which some event has already occurred. From this point of view, the formula for conditional probability becomes natural, and it leads to independence, dependence, partitions, the law of total probability, and Bayes' theorem. Bayes' theorem is treated as a central tool for reversing conditional reasoning and for understanding situations where intuition is easily misled.

The course then introduces random variables through the discrete case. A random variable appears as a function that extracts a numerical quantity from an elementary outcome: the number of successes, the number of errors, the number of arrivals, a payoff, or another measurable feature of the experiment. This leads to the support of a discrete random variable, its probability mass function, its cumulative distribution function, expectation, variance, moments, and interpretation. Classical discrete laws are presented as models arising from specific mechanisms, not as isolated formulas.

The continuous case is then developed as a parallel but conceptually more demanding part of the theory. For a continuous random variable, probability is no longer concentrated at individual points, so probabilities of intervals become the fundamental objects. This leads to density functions, cumulative distribution functions, integrals, continuous expectations, variances, moments, quantiles, and the interpretation of probability as area under a density. Important continuous models, including the uniform, exponential, normal, gamma, and beta distributions, are introduced with attention to their meaning and use.

Once theoretical probability laws have been introduced, the course moves to empirical distributions and descriptive statistics. The same ideas now appear in data: frequency tables, histograms, empirical cumulative distribution functions, means, medians, variances, standard deviations, quartiles, interquartile ranges, and outliers. This stage is meant to clarify the difference between an idealized probability law and a finite dataset, and also to show how empirical summaries mirror the theoretical quantities defined earlier.

The following stage adds another level of randomness by studying sampling and distributions of statistics. A sample mean, sample proportion, sample variance, median, or other statistic is not merely a number; it is itself a random variable because it depends on the random sample. This perspective prepares the ground for understanding sampling distributions and for seeing why repeated sampling creates a new layer of probabilistic structure.

The final stage of the present course outline is devoted to limit theorems and statistical inference. Once empirical summaries and sampling distributions are understood, we can ask what regularities appear as the sample size grows or as experiments are repeated many times. This leads to the law of large numbers, the central limit theorem, normal approximations, standard errors, confidence intervals, and the basic logic of statistical inference. The goal is to show how probability theory makes inference from data possible: it allows us to estimate unknown quantities, measure uncertainty, and justify conclusions mathematically.

Chapter 2

Elements of Combinatorics

2.1 Counting begins with elementary outcomes

Combinatorics is often presented as a list of formulas. In probability this is dangerous. Before a formula can be used, we must know what one elementary outcome is. A physical experiment does not determine a sample space by itself. The sample space depends on what is recorded.

Suppose that three balls are drawn from a box. If the balls are labelled and the order of drawing is recorded, then an outcome is a sequence of labelled objects. If only the selected balls are recorded, then an outcome is a set. If only the colors are recorded, then labelled balls of the same color may become indistinguishable. If only the number of red balls is recorded, then the outcome is no longer the full draw, but only the value of a random variable.

Thus the first question in a counting problem is not

Which formula should be used?

The first question is

What is one outcome?

Definition

An **elementary outcome** is a complete description of one possible result of an experiment, according to the recording rule chosen for the model.

The phrase "according to the recording rule" is essential. The same physical situation may produce different mathematical sample spaces if different information is recorded.

Example

A box contains three balls labelled R_1, R_2, B_1. Two balls are drawn without replacement. If order and labels are recorded, then examples of outcomes are

$$(R_1, B_1), \qquad (B_1, R_1), \qquad (R_1, R_2).$$

If only the set of labelled balls is recorded, then examples of outcomes are

$$\{R_1, B_1\}, \qquad \{R_1, R_2\}.$$

If only the sequence of colors is recorded, then examples of outcomes are

$$(R, B), \qquad (B, R), \qquad (R, R).$$

If only the number of red balls is recorded, then the possible recorded values are

$$0, 1, 2.$$

These are not four different answers to the same counting question. They are four different mathematical models of the same physical experiment.

Remark

A counting error often begins before any arithmetic is done. It begins when the outcome type is chosen incorrectly.

2.2 Recording rules and sample spaces

A recording rule tells us which differences between physical results are mathematically visible. Some differences are recorded; others are ignored. Once the recording rule is fixed, the sample space is the set of all possible recorded outcomes.

Definition

A **recording rule** is a decision about what information is included in the mathematical outcome and what information is ignored.

For finite experiments, the construction usually follows this pattern:

$$\text{physical situation} \longrightarrow \text{recording rule} \longrightarrow \text{elementary outcome} \longrightarrow \text{sample space.}$$

The following table shows common outcome types.

Recording rule	Outcome type	Typical notation
Order is recorded	sequence	$(a_1, a_2, \dots, a_k)$
Only membership is recorded	set	$\{a_1, a_2, \dots, a_k\}$
All objects are arranged in a row	linear arrangement	$(a_{\sigma(1)}, \dots, a_{\sigma(n)})$
Objects are arranged on a circle	circular arrangement	relative order
Only numbers of types are recorded	count vector	$(n_1, n_2, \dots, n_r)$
Only a numerical summary is recorded	random variable value	$X(\omega)$

Example

A committee of three people is chosen from ten people. If the committee is recorded only as a group, then

$$\{A, B, C\} = \{C, A, B\}.$$

The order of listing the names is irrelevant. The outcome is a set.
If the same three people are assigned the roles of chair, secretary, and treasurer, then

$$(A, B, C) \neq (C, A, B).$$

The outcome is now an ordered assignment of roles.

The two situations may look similar in everyday language: in both cases three people are selected. But mathematically they are different because the recording rules are different.

2.3 Diagnostic questions before counting

Before using a formula, answer the following questions.

1. What is one elementary outcome?

2. Is the outcome a sequence, a set, an arrangement, or a count vector?

3. Does order matter?
4. Are all objects used, or only some of them?
5. Can the same object be used more than once?
6. Are the objects distinguishable or indistinguishable?
7. Are we first counting labelled descriptions and then identifying descriptions that represent the same outcome?

These questions are more important than the names of formulas. The names are only labels attached to structures that have already been recognized.

Remark

Do not begin a problem by asking which formula you remember. Begin by asking what kind of outcome is being constructed.

2.4 The multiplication principle as a construction rule

Many counting problems can be solved by describing how an outcome is constructed step by step.

Theorem

Multiplication principle. Suppose a construction is performed in k successive stages. If there are m_1 choices for the first stage, m_2 choices for the second stage, and so on up to m_k choices for the k-th stage, then the number of possible complete constructions is

$$m_1 m_2 \cdots m_k.$$

The multiplication principle is not only a formula. It is a way of reading an outcome as a construction process.

Example

A four-digit PIN code is formed from the digits $0, 1, \ldots, 9$, and digits may repeat. One outcome is a sequence such as

$$(0, 7, 7, 3).$$

There are 10 choices for each position, hence

$$10 \cdot 10 \cdot 10 \cdot 10 = 10^4$$

possible PIN codes.
The reason is not that the formula n^k was memorized. The reason is that a PIN code is a sequence of four independent choices from a set of ten symbols.

Example

A three-letter code is formed from the letters A, B, C, D, and letters may not repeat. One outcome is an ordered triple, for example

$$(A, D, B).$$

There are 4 choices for the first position, 3 for the second, and 2 for the third. Therefore the number of codes is

$$4 \cdot 3 \cdot 2 = 24.$$

When the same number of choices is available at every stage, powers appear. When the number of choices decreases, falling products appear. When order is later ignored, division appears. Most elementary combinatorial formulas can be understood from these three ideas.

2.5 Sequences with repetition

A sequence with repetition is an ordered list in which each position is filled independently from the same set of symbols.

Definition

Let S be a set with n elements. A **sequence of length k with repetition** over S is an ordered k-tuple

$$(a_1, a_2, \ldots, a_k),$$

where each $a_i \in S$. The same element may occur many times.

The number of such sequences is

$$n^k.$$

This model is used when:

- order is recorded,
- each position is chosen separately,
- repetition is allowed,
- the outcome is a full sequence.

Typical examples include PIN codes, repeated coin tosses, repeated die rolls, strings over an alphabet, and repeated selections with replacement.

Example

The sample space for three tosses of a coin is

$$\Omega = \{H, T\}^3.$$

It has

$$2^3 = 8$$

elementary outcomes. Each outcome is a sequence, for example

$$(H, T, H).$$

The outcomes (H, T, H) and (T, H, H) are different because the order of tosses is recorded.

2.6 Ordered selections without repetition

Sometimes only some objects are chosen, but the order or position of the chosen objects matters.

Definition

An **ordered selection without repetition** of length k from n distinct objects is an ordered list of k different objects chosen from the n available objects.

The number of ordered selections without repetition is

$$P(n,k) = n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}.$$

This model is used when:

- only some objects are selected,
- order or position matters,
- no object may be used twice.

Example

Gold, silver, and bronze medals are assigned among 12 runners. An outcome is not a set of three runners. It is an ordered triple

$$(\text{gold}, \text{silver}, \text{bronze}).$$

There are

$$12 \cdot 11 \cdot 10 = P(12,3)$$

possible assignments.

In some European terminology, ordered selections are called variations. In these notes we use the name ordered selections without repetition, or k-permutations, because it describes directly what the outcome is.

2.7 Arrangements of all objects

A permutation is an arrangement of all objects in a row.

Definition

A **permutation** of n distinct objects is a linear arrangement of all n objects.

There are

$$n!$$

permutations of n distinct objects.

This is a special case of ordered selection without repetition with $k = n$:

$$P(n,n) = n!.$$

Example

If 7 students are arranged in a line, then one outcome is a complete ordered list of all students. The number of arrangements is

$$7!.$$

Permutations are used when:

- all objects are used,
- order matters,
- the objects are distinguishable,
- the arrangement is linear.

2.8 Circular arrangements

A circular arrangement is different from a linear arrangement because rotating the whole arrangement does not create a new relative seating order.

Example

Suppose four people A, B, C, D sit around a round table. The linear lists

$$(A, B, C, D), \qquad (B, C, D, A), \qquad (C, D, A, B), \qquad (D, A, B, C)$$

represent the same circular arrangement if only relative positions matter.

For n distinct objects placed around a circle, with rotations identified, the number of circular arrangements is

$$(n-1)!.$$

One way to see this is to fix one object as a reference point. The remaining $n-1$ objects can then be arranged around it in

$$(n-1)!$$

ways.

Remark

A circular arrangement requires a modelling decision. If seats are numbered or if there is a distinguished position, then rotations may become different outcomes and the linear count may be appropriate.

2.9 Unordered selections: combinations

A combination is an unordered selection.

Definition

A **combination** of size k from an n-element set is a subset with k elements.

The number of combinations is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

This formula can be derived from ordered selections. There are

$$P(n, k) = \frac{n!}{(n-k)!}$$

ordered selections of length k. But each unordered selection of k objects can be ordered in $k!$ ways. Therefore

$$\binom{n}{k} = \frac{P(n,k)}{k!} = \frac{n!}{k!(n-k)!}.$$

Combinations are used when:

- only membership is recorded,
- order is ignored,
- each object may be chosen at most once,
- the outcome is a set.

Example

A five-card hand from a standard deck is a set of five cards, not a sequence of five draws, if the final hand is all that is recorded. The number of possible hands is

$$\binom{52}{5}.$$

If the order of drawing is recorded, the correct count is instead

$$P(52,5) = 52 \cdot 51 \cdot 50 \cdot 49 \cdot 48.$$

These are different sample spaces.

2.10 Overcounting and equivalent descriptions

Many combinatorial formulas arise from a two-step idea:

1. Count descriptions that are easy to construct.
2. Divide by the number of descriptions representing the same outcome.

This is the idea of overcounting.

Definition

Overcounting occurs when the same intended outcome is counted more than once because it has several different descriptions.

Overcounting is not a mistake if it is controlled. It becomes a method when we know exactly how many times each outcome has been counted.

Example

Suppose we want to choose a committee of three people from n people. The order of names in the committee does not matter.
It is easier first to count ordered selections:

$$n(n-1)(n-2).$$

But each committee of three people has been counted $3!$ times, because the same three people can be listed in 3! different orders. Therefore the number of committees is

$$\frac{n(n-1)(n-2)}{3!}.$$

The essential idea is not division itself. The essential idea is identifying which descriptions should be treated as the same outcome.

2.11 Distinguishable and indistinguishable objects

Objects are distinguishable if the model treats them as separate individuals. Objects are indistinguishable if only their type matters.

Example

The objects

$$R_1, R_2, R_3$$

are three distinguishable red balls. The symbols

$$R, R, R$$

represent three indistinguishable red balls of the same type.

A physical object may be distinguishable in one model and indistinguishable in another. This depends on the recording rule.

Example

An urn contains red and blue balls. If each ball has a label, and labels are recorded, then two red balls are different outcomes. If only colors are recorded, then the red balls are treated as indistinguishable.

This distinction is crucial in probability. If the physical random mechanism selects labelled balls uniformly, but the sample space records only colors, then the color outcomes may not be equally likely.

2.12 Permutations with repeated elements

When some objects are indistinguishable, a linear arrangement may have several labelled descriptions that represent the same visible arrangement.

Suppose we arrange n objects, where there are n_1 objects of type 1, n_2 objects of type 2, and so on, with

$$n_1 + n_2 + \cdots + n_r = n.$$

The number of distinct visible arrangements is

$$\frac{n!}{n_1! n_2! \cdots n_r!}.$$

Example

The word BANANA contains six letters: three A's, two N's, and one B. If all six positions are arranged, then the number of distinct words is

$$\frac{6!}{3!2!1!}.$$

The division appears because permuting the three A's among themselves does not change the visible word, and neither does permuting the two N's.

This formula should be read as an overcounting correction:

$$\text{labelled arrangements} \quad \longrightarrow \quad \text{visible arrangements.}$$

2.13 Combinations with repetition

Sometimes we choose k objects from n types, and the same type may be chosen several times. The order of the chosen types is ignored. The outcome is then not a sequence but a vector of counts.

Definition

A **combination with repetition** of size k from n types is a choice of non-negative integers

$$(x_1, x_2, \ldots, x_n)$$

such that

$$x_1 + x_2 + \cdots + x_n = k.$$

Here x_i records how many objects of type i are chosen.

The number of such choices is

$$\binom{n+k-1}{k} = \binom{n+k-1}{n-1}.$$

This is often called the stars and bars formula. The idea is to represent the k chosen objects as stars and use $n-1$ bars to separate the counts of the n types.

Example

How many ways are there to choose 5 donuts from 3 flavors if only the numbers of each flavor matter?
An outcome is a count vector

$$(x_1, x_2, x_3), \qquad x_1 + x_2 + x_3 = 5.$$

The number of outcomes is

$$\binom{3+5-1}{5} = \binom{7}{5} = \binom{7}{2}.$$

Remark

Combinations with repetition do not describe ordered repeated choices. If order matters, the model is a sequence with repetition and the count is n^k. If order is ignored and only counts of types matter, the model is combinations with repetition.

2.14 Counting favorable events in probability

Combinatorics becomes part of probability when it is used to count sample spaces and events.

If Ω is finite and all elementary outcomes are equally likely, then

$$P(A) = \frac{|A|}{|\Omega|}.$$

This formula is valid only after the sample space has been correctly chosen and the elementary outcomes are equally likely.

Example

A fair die is rolled twice. If order is recorded, then

$$\Omega = \{1,2,3,4,5,6\}^2, \qquad |\Omega| = 36.$$

The event that the sum is 7 is

$$A = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}.$$

Thus

$$P(A) = \frac{6}{36} = \frac{1}{6}.$$

Example

Five cards are drawn from a standard deck, and the final hand is recorded. If every five-card hand is equally likely, then

$$|\Omega| = \binom{52}{5}.$$

The number of hands with exactly two hearts is

$$\binom{13}{2}\binom{39}{3}.$$

Therefore

$$P(\text{exactly two hearts}) = \frac{\binom{13}{2}\binom{39}{3}}{\binom{52}{5}}.$$

The important point is that the denominator and numerator must be counted inside the same model. It is wrong to count ordered outcomes in the denominator and unordered outcomes in the numerator, or conversely.

2.15 Changing the recording rule changes the model

The same experiment may be counted in different ways because different questions require different sample spaces.

Example

Three balls are drawn without replacement from an urn containing labelled balls. Compare the following recording rules.

Recorded information	Outcome type	Typical count
labels in order	sequence without repetition	$P(n, 3)$
labels only, order ignored	subset	$\binom{n}{3}$
colors in order	sequence of colors	depends on color counts
color counts only	count vector	depends on available colors
number of red balls only	value of a random variable	$0, 1, 2, 3$ possible

The physical action is the same: three balls are drawn. The mathematical model is not the same, because the recorded outcome is different.

2.16 Recognizing the correct counting model

The following diagnostic table summarizes the main finite counting structures.

Outcome type	Order?	Repetition?	Count
sequence with repetition	yes	yes	n^k
ordered selection	yes	no	$P(n, k) = \dfrac{n!}{(n-k)!}$
permutation	yes	no, all used	$n!$
circular arrangement	relative order	no, all used	$(n-1)!$
combination	no	no	$\binom{n}{k}$
combination with repetition	no	yes	$\binom{n+k-1}{k}$
multiset permutation	positions yes	identical objects	$\dfrac{n!}{n_1! \cdots n_r!}$

The table should be used only after the outcome type has been identified. It is a summary, not a substitute for modelling.

Remark

A formula is a compressed argument. If the argument does not match the situation, the formula gives the wrong count even when the arithmetic is correct.

2.17 Common traps

Trap 1: confusing a sequence with a set

The ordered outcomes

$$(A, B, C) \quad \text{and} \quad (C, B, A)$$

are different sequences but the same set:

$$\{A, B, C\} = \{C, B, A\}.$$

Use sequences when order is recorded. Use sets when only membership is recorded.

Trap 2: treating indistinguishable objects as distinguishable

The letters in BANANA are not six distinct letters. The three A's are indistinguishable, and the two N's are indistinguishable. Therefore $6!$ overcounts the number of visible arrangements.

Trap 3: using combinations when roles matter

A committee is a set. A medal assignment is an ordered assignment. A group of three athletes and a gold-silver-bronze result are not the same kind of outcome.

Trap 4: assuming all recorded outcomes are equally likely

If labelled balls are selected uniformly but only colors are recorded, then color outcomes may have different probabilities. A smaller recorded sample space does not automatically become equally likely.

Trap 5: mixing models in one probability computation

In the formula

$$P(A) = \frac{|A|}{|\Omega|},$$

both $|A|$ and $|\Omega|$ must be counted using the same outcome type. Do not count ordered draws in one part and unordered hands in the other.

2.18 From combinatorics to probability models

The role of combinatorics in this course is not to memorize more formulas. Its role is to build sample spaces and events precisely.

The general workflow is

$$\text{situation} \longrightarrow \text{recording rule} \longrightarrow \text{elementary outcome} \longrightarrow \text{sample space} \longrightarrow \text{event} \longrightarrow \text{probability}.$$

In later chapters, many probability distributions will arise by recording only a numerical summary of a richer experiment. For example, in n Bernoulli trials the full elementary outcome may be a sequence such as

$$(S, F, S, S, F),$$

while the random variable records only the number of successes:

$$X = 3.$$

The binomial coefficient appears because many different sequences have the same number of successes.

Thus combinatorics is the bridge between concrete experiments and probability laws. It explains how many elementary outcomes produce the same event, the same count, or the same value of a random variable.

2.19 Final checklist

When solving a counting problem, write down the following before doing arithmetic:

1. What is the physical situation?
2. What information is recorded?
3. What is one elementary outcome?
4. What is the sample space?
5. Are outcomes sequences, sets, arrangements, or count vectors?

6. Are the elementary outcomes equally likely if probability is being computed?
7. Which construction or overcounting argument gives the count?
8. What does the final number mean in the original situation?

This checklist is the main lesson of the chapter. The formulas are useful, but they are useful only when the underlying structure has been discovered first.

Chapter 3

Discovering Formalism

This chapter develops the conceptual language from which probability theory begins. The aim is to show how informal statements about a random situation are transformed into mathematical objects: elementary outcomes, events, operations on events, and finally probability as a numerical structure placed on a family of events.

The chapter is not intended as a collection of computational techniques. Its role is to train careful analysis, abstraction, and the construction of formal language. Students should learn to see that statements about an experiment select sets of outcomes, and that logical relations between statements become set-theoretic relations between events.

Chapter 4

Sample Spaces and Elementary Probability

This chapter develops several canonical examples of sample spaces and elementary probability from the ground up. The emphasis is on understanding the experiment, deciding what should count as an elementary outcome, constructing the sample space, defining events, and assigning probabilities only after the mathematical model has been made explicit.

Different representations of the same probabilistic structure should be used throughout the chapter: lists, tables, trees, diagrams, and symbolic notation. The goal is to show that a sample space is an abstract object produced by analysis and modelling, not a collection of observed data.

Chapter 5

Conditional Probability, Independence, and Bayes' Theorem

This chapter explores the algebra of events and the concept of conditional probability, which are fundamental to understanding how probabilities are updated with new information.

Chapter 6

Discrete Random Variables and Probability Laws

This chapter introduces random variables through the discrete case. A random variable is viewed as a function that extracts a numerical quantity from an elementary outcome and allows probabilistic phenomena to be studied through numerical characteristics.

The chapter develops discrete probability laws, probability mass functions, cumulative distribution functions, expectation, variance, moments, and the interpretation of these quantities. Classical discrete models are presented as mathematical descriptions of specific random mechanisms.

Chapter 7

Continuous Random Variables and Probability Laws

This chapter develops the continuous counterpart of discrete probability laws. The central conceptual change is that probability is no longer concentrated at individual points; instead, probabilities of intervals become the fundamental objects.

The chapter introduces probability density functions, cumulative distribution functions, integrals, expectation, variance, moments, quantiles, and the interpretation of probability as area under a density. Continuous models such as the uniform, exponential, normal, gamma, and beta distributions are treated as tools for modelling continuous random phenomena.

Chapter 8

Empirical Distributions and Descriptive Statistics

This chapter studies empirical distributions and descriptive statistics obtained from data. The concepts developed earlier for theoretical probability laws are now examined through observed datasets.

Frequency tables, histograms, empirical distribution functions, means, medians, variances, standard deviations, quantiles, and related descriptive measures are introduced as tools for describing and understanding data.

Chapter 9

Sampling and Distributions of Statistics

This chapter introduces the idea that statistics computed from samples are themselves random variables. Sample means, proportions, variances, medians, and other summaries acquire their own probability laws when sampling is repeated.

The purpose of the chapter is to prepare the transition from descriptive statistics to statistical inference through the study of sampling distributions.

Chapter 10

Limit Theorems and Statistical Inference

This chapter studies the regularities that emerge when sample sizes become large and experiments are repeated many times. The law of large numbers and the central limit theorem provide the mathematical foundation for statistical inference.

The chapter introduces normal approximations, standard errors, confidence intervals, and the basic principles of inference from data.